Safety data sheet

1. Substance/preparation and manufacturer/supplier identification

Formic acid 94%

Use: Chemical used in synthesis and/or formulation of industrial products

Manufacturer/supplier:
BASF-YPC Company Limited
No.266 Ethylene Rd,
Luhe District, Nanjing, Jiangsu
Postal Code:210048, CHINA
Telephone: +86 25 5856-2402
Telefax number: +86 25 5856-9077
E-mail address: bycerc@basf-ypc.com.cn

Emergency information:
International emergency number:
Telephone: +49 180 2273-112

2. Hazard identification

Classification according to UN GHS 2009

Classification of the substance and mixture:
Flammable liquids: Cat. 3
Acute toxicity: Cat. 3 (Inhalation - vapour)
Acute toxicity: Cat. 4 (oral)
Skin corrosion/irritation: Cat. 1A
Serious eye damage/eye irritation: Cat. 1

Label elements and precautionary statement:

Pictogram:

Signal Word:
Danger

Hazard Statement:

H226	Flammable liquid and vapour.
H331	Toxic if inhaled.
H302	Harmful if swallowed.
H314	Causes severe skin burns and eye damage.

Precautionary Statements (Prevention):

P271	Use only outdoors or in a well-ventilated area.
P280	Wear protective gloves, protective clothing and eye protection or face protection.
P210	Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
P260	Do not breathe mist or vapour.
P243	Take action to prevent static discharges.
P241	Use explosion-proof electrical, ventilating and lighting equipment.
P264	Wash contaminated body parts thoroughly after handling.
P270	Do not eat, drink or smoke when using this product.
P242	Use only non-sparking tools.
P240	Ground and bond container and receiving equipment.

Precautionary Statements (Response):

P310	Immediately call a POISON CENTER or physician.
P305 + P351 + P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P304 + P340	IF INHALED: Remove person to fresh air and keep comfortable for breathing.
P303 + P361 + P352	IF ON SKIN (or hair): Remove/Take off immediately all contaminated clothing. Wash with plenty of soap and water.
P301 + P330 + P331	IF SWALLOWED: rinse mouth. Do NOT induce vomiting.
P370 + P378	In case of fire: Use alcohol-resistant foam, carbon dioxide, dry powder or water spray for extinction.

Precautionary Statements (Storage):

P403 + P235	Store in a well-ventilated place. Keep cool.
P233	Keep container tightly closed.
P405	Store locked up.

Precautionary Statements (Disposal):

P501	Dispose of contents and container to hazardous or special waste collection point.

Other hazards which do not result in classification:
If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.

Corrosive to the respiratory tract.

3. Composition/information on ingredients

Chemical nature

formic acid (Content (W/W): 94 %)
CAS Number: 64-18-6
water (Content (W/W): 6 %)
CAS Number: 7732-18-5

Hazardous ingredients

formic acid

Content (W/W): 94 % CAS Number: 64-18-6	Flam. Liq.: Cat. 3 Acute Tox.: Cat. 3 (Inhalation - vapour) Acute Tox.: Cat. 4 (oral) Skin Corr./Irrit.: Cat. 1A Eye Dam./Irrit.: Cat. 1

4. First-Aid Measures

General advice:
First aid personnel should pay attention to their own safety. If the patient is likely to become unconscious, place and transport in stable sideways position (recovery position). Immediately remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention. Immediately administer a corticosteroid from a controlled/metered dose inhaler.

On skin contact:
Immediately wash thoroughly with plenty of water, apply sterile dressings, consult a skin specialist.

On contact with eyes:
Immediately wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:
Do not induce vomiting. Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

Note to physician:
Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11.
Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures

Suitable extinguishing media:

water spray, dry powder, alcohol-resistant foam, carbon dioxide

Specific hazards:
carbon monoxide
The substances/groups of substances mentioned can be released if the product is involved in a fire.

Special protective equipment:
Wear self-contained breathing apparatus and chemical-protective clothing.

Further information:
Collect contaminated extinguishing water separately, do not allow to reach sewage or effluent systems.

6. Accidental Release Measures

Personal precautions:
Avoid contact with the skin, eyes and clothing. Breathing protection required.

Environmental precautions:
Do not empty into drains.

Methods for cleaning up or taking up:
For large amounts: Pump off product.
For residues: Pick up with suitable absorbent material (e.g. acid binder). Dispose of absorbed material in accordance with regulations.

7. Handling and Storage

Handling

Ensure thorough ventilation of stores and work areas. Sealed containers should be protected against heat as this results in pressure build-up.

Protection against fire and explosion:
Sources of ignition should be kept well clear.

Storage

Segregate from alkalies and alkalizing substances.

Suitable materials for containers: Stainless steel 1.4571, Stainless steel 1.4404, High density polyethylene (HDPE), Low density polyethylene (LDPE), glass

Storage stability:
Storage temperature: < 30 °C
Storage duration: 36 Months
From the data on storage duration in this safety data sheet no agreed statement regarding the warrantee of application properties can be deduced.

8. Exposure controls and personal protection

Components with occupational exposure limits

formic acid, 64-18-6;
TWA value 5 ppm (ACGIHTLV)
STEL value 10 ppm (ACGIHTLV)
TWA value 5 ppm (OEL (TH))

Personal protective equipment

Respiratory protection:
Suitable respiratory protection for lower concentrations or short-term effect: Gas filter for acid inorganic gases/vapours such as SO2, HCl (e.g. EN 14387 Type E). Gas filter for gases/vapours of inorganic compounds (e.g. EN 14387 Type B) Combination filter for gases/vapours of organic, inorganic, acid inorganic and alkaline compounds (e.g. EN 14387 Type ABEK). Suitable respiratory protection for higher concentrations or long-term effect: Self-contained breathing apparatus.

Hand protection:
Chemical resistant protective gloves (EN 374)
Suitable materials also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN 374):
chloroprene rubber (CR) - 0.5 mm coating thickness
butyl rubber (butyl) - 0.7 mm coating thickness
fluoroelastomer (FKM) - 0.7 mm coating thickness
Polyethylene-Laminate (PE laminate) - ca. 0.1 mm coating thickness
Suitable materials for short-term contact (recommended: At least protective index 2, corresponding > 30 minutes of permeation time according to EN 374)
polyvinylchloride (PVC) - 0.7 mm coating thickness
natural rubber/natural latex (NR) - 0.5 mm coating thickness
Manufacturer's directions for use should be observed because of great diversity of types.
Supplementary note: The specifications are based on tests, literature data and information of glove manufacturers or are derived from similar substances by analogy. Due to many conditions (e.g. temperature) it must be considered, that the practical usage of a chemical-protective glove in practice may be much shorter than the permeation time determined through testing.

Eye protection:
Tightly fitting safety goggles (cage goggles) (e.g. EN 166) and face shield.

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures:
Avoid contact with the skin, eyes and clothing. Avoid inhalation of vapour. Avoid contact with skin and eyes. Gloves must be inspected regularly and prior to each use. Replace if necessary (e.g. pinhole leaks). Take off immediately all contaminated clothing. Wash contaminated clothing before reuse. Hands and/or face should be washed before breaks and at the end of the shift. When using, do not eat, drink or smoke.

9. Physical and Chemical Properties

Form:	liquid	
Colour:	colourless to yellow	
Odour:	pungent odour	
Odour threshold:	not determined	
pH value:	2.2 (10 g/l, 20 °C)	
pKA:	3.70 (20 °C)	(OECD Guideline 112)
Melting point:	-2 °C	
Boiling point:	103 °C	
Flash point:	56 °C	(DIN 51755)
Evaporation rate:	Value can be approximated from Henry's Law Constant or vapor pressure.	
Flammability (solid/gas):	Flammable liquid and vapour.	(derived from flash point)
Lower explosion limit:	For liquids not relevant for classification and labelling., The lower explosion point may be 5 - 15 °C below the flash point.	
Upper explosion limit:	For liquids not relevant for classification and labelling.	
Ignition temperature:	480 °C	(DIN 51794)
Thermal decomposition:	350 °C , 0.15 kJ/g Thermal decomposition above the indicated temperature is possible. It is not a self-decompositionable substance.	(DSC (DIN 51007))
Self ignition:	Based on its structural properties the product is not classified as self-igniting.	Test type: Spontaneous self-ignition at room-temperature.
Self heating ability:	not applicable, the product is a liquid	
Explosion hazard:	Based on the chemical structure there is no indication of explosive properties.	
Fire promoting properties:	Based on its structural properties the product is not classified as oxidizing.	
Vapour pressure:	36 mbar (20 °C) 148 mbar (50 °C)	

Density: 1.2116 g/cm3
(20 °C)
1.1874 g/cm3
(40 °C)
1.1691 g/cm3
(55 °C)
1.2200 g/cm3
(15 °C)
1.1800 g/cm3
(50 °C)

Relative density: 1.2195 (OECD Guideline 109)
(20 °C)

Solubility in water: miscible
(20 °C, 1,013.25 hPa)

Miscibility with water:
miscible in all proportions

Solubility (qualitative) solvent(s): organic solvents
miscible

Partitioning coefficient n-octanol/water (log Pow): -2.1 (Directive 92/69/EEC, A.8)
(23 °C; pH value: 7.0)
-1.9 (Directive 92/69/EEC, A.8)
(23 °C; pH value: 5.0)
-2.3 (Directive 92/69/EEC, A.8)
(23 °C; pH value: 9.0)

Adsorption/water - soil: KOC: < 17.8; log KOC: 1.25 (OECD Guideline 121)

Surface tension: 71.5 mN/m (OECD-Guideline 115)
(20 °C; 1 g/l)

Viscosity, dynamic: 1.78 mPa.s (calculated (from kinematic viscosity))
(20 °C)
1.21 mPa.s (calculated (from kinematic viscosity))
(40 °C)
0.95 mPa.s (calculated (from kinematic viscosity))
(55 °C)

Viscosity, kinematic: 1.47 mm2/s (DIN 51562)
(20 °C)
1.02 mm2/s (DIN 51562)
(40 °C)
0.81 mm2/s (DIN 51562)
(55 °C)

Molar mass: 46.03 g/mol

10. Stability and Reactivity

Conditions to avoid:
Temperature: > 30 °C

Thermal decomposition: 350 °C, 0.15 kJ/g (DSC (DIN 51007))

Thermal decomposition above the indicated temperature is possible. It is not a self-decompositionable substance.

Substances to avoid:
bases, non-coated metals, base metals

Corrosion to metals: No corrosive effect on metal.

Corrosion to metals: No corrosive effect on metal.

Hazardous reactions:
Exothermic reaction. Reacts with alkalies. Reacts with amines.

Possible thermal decomposition products:
carbon monoxide

11. Toxicological Information

Acute toxicity

Assessment of acute toxicity:
Of moderate toxicity after single ingestion. Of pronounced toxicity after short-term inhalation.

Experimental/calculated data:
LD50 rat (oral): 730 mg/kg (OECD Guideline 401)

LC50 rat (by inhalation): 7.85 mg/l 4 h (BASF-Test)
The vapour was tested.

(dermal):No data available. Study scientifically not justified.

Irritation

Assessment of irritating effects:
Highly corrosive! Damages skin and eyes.

Experimental/calculated data:
Skin corrosion/irritation rabbit: Corrosive. (OECD Guideline 404)
Literature data.

Serious eye damage/irritation:Study scientifically not justified. As the product corrodes the skin, it can be expected to have a similar effect on the eyes also.

Respiratory/Skin sensitization

Assessment of sensitization:
Skin sensitizing effects were not observed in animal studies.

Experimental/calculated data:
Buehler test guinea pig: Non-sensitizing. (OECD Guideline 406)

Germ cell mutagenicity

Assessment of mutagenicity:
No mutagenic effect was found in various tests with bacteria and mammalian cell culture. The substance was not mutagenic in an insect test.

Carcinogenicity

Assessment of carcinogenicity:
In long-term studies in rats and mice in which the substance was given by feed, a carcinogenic effect was not observed. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Reproductive toxicity

Assessment of reproduction toxicity:
The results of animal studies gave no indication of a fertility impairing effect. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Developmental toxicity

Assessment of teratogenicity:
No indications of a developmental toxic / teratogenic effect were seen in animal studies. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Specific target organ toxicity (single exposure):

Assessment of STOT single:
Corrosive to the respiratory tract.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
No substance-specific organtoxicity was observed after repeated administration to animals. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Aspiration hazard

No aspiration hazard expected.

12. Ecological Information

Ecotoxicity

Assessment of aquatic toxicity:
There is a high probability that the product is not acutely harmful to aquatic organisms. The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations.

The product gives rise to pH shifts.

Toxicity to fish:
LC50 (96 h) 130 mg/l, Brachydanio rerio (OECD 203; ISO 7346; 92/69/EEC, C.1, static)
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Aquatic invertebrates:
EC50 (48 h) 365 mg/l, Daphnia magna (OECD Guideline 202, part 1, static)
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition. The statement of the toxic effect relates to the analytically determined concentration.

Aquatic plants:
EC50 (72 h) 1,240 mg/l (growth rate), Selenastrum capricornutum (OECD Guideline 201, static)
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

EC50 (72 h) 32.64 mg/l (growth rate), Scenedesmus subspicatus (DIN 38412 Part 9, static)
The details of the toxic effect relate to the nominal concentration. The product will cause changes in the pH value of the test system. The result refers to an unneutralized sample.

Microorganisms/Effect on activated sludge:
EC10 (13 d) 72 mg/l, activated sludge, domestic, non-adapted (other, aerobic)

Chronic toxicity to fish:
Study scientifically not justified.

Chronic toxicity to aquatic invertebrates:
No observed effect concentration (21 d), >= 100 mg/l, Daphnia magna (OECD Guideline 211, semistatic)
The statement of the toxic effect relates to the analytically determined concentration. The product will cause changes in the pH value of the test system. The result refers to a neutralized sample. No effects at the highest test concentration.

Assessment of terrestrial toxicity:
Study scientifically not justified.

Soil living organisms:
Literature data.

Terrestrial plants:
Literature data.

Other terrestrial non-mammals:
LD50 (18 h) >= 111 mg/kg, Agelaius phoeniceus
Literature data.

Mobility

Assessment transport between environmental compartments:
The substance will not evaporate into the atmosphere from the water surface.
Adsorption to solid soil phase is not expected.

Persistence and degradability

Elimination information:
100 % DOC reduction (9 d) (OECD 301E/92/69/EEC, C.4-B) (aerobic, municipal sewage treatment plant effluent)

Assessment of stability in water:
According to structural properties, hydrolysis is not expected/probable.

Information on Stability in Water (Hydrolysis):
$t_{1/2}$ > 5 d (50 °C, pH value 4), (Directive 92/69/EEC, C.7, pH 4)

$t_{1/2}$ > 5 d (50 °C, pH value 7), (Directive 92/69/EEC, C.7, pH 7)

$t_{1/2}$ > 5 d (50 °C, pH value 9), (Directive 92/69/EEC, C.7, pH 9)

Sum parameter

Chemical oxygen demand (COD): 348 mg/g

Biochemical oxygen demand (BOD) Incubation period 5 d: 86 mg/g

Bioaccumulation potential

Assessment bioaccumulation potential:
Significant accumulation in organisms is not to be expected.

Bioaccumulation potential:
Significant accumulation in organisms is not to be expected.

13. Disposal Considerations

Incinerate in suitable incineration plant, observing local authority regulations.
A waste code in accordance with the European waste catalog (EWC) cannot be specified, due to dependence on the usage.
The waste code in accordance with the European waste catalog (EWC) must be specified in cooperation with disposal agency/manufacturer/authorities.

Contaminated packaging:
Contaminated packaging should be emptied as far as possible; then it can be passed on for recycling after being thoroughly cleaned.

14. Transport Information

Domestic transport:

Packing group:	II
ID number:	UN 1779
Transport hazard class(es):	8, 3
Proper shipping name:	FORMIC ACID

Sea transport
IMDG

Packing group:	II
ID number:	UN 1779
Transport hazard class(es):	8, 3
Marine pollutant:	NO
Proper shipping name:	FORMIC ACID

Air transport
IATA/ICAO

Packing group:	II
ID number:	UN 1779
Transport hazard class(es):	8, 3
Proper shipping name:	FORMIC ACID

15. Regulatory Information

Hazard determining component(s) for labelling: FORMIC ACID

Other regulations

as in Annex I of Directive 67/548/EEC

16. Other Information

flue gas desulphurization rubber industry textile industry leather industry

Vertical lines in the left hand margin indicate an amendment from the previous version.